

Resume: Jacob Haflett

Senior Backend and Data Engineer | Data Platforms, Distributed Systems, AI Pipelines

+1 859-361-0084 • jacobhaflett@icloud.com • haflettjm.github.io • github.com/haflettjm

SUMMARY

Backend and data engineer with 5+ years building data-intensive platforms from idea to stable production. Scaled Spark pipelines across catalogs of millions of SKUs and a 35,000-robot fleet, shipped multi-tenant APIs handling 100K requests/sec at 99.9%+ availability, and built integrations with external systems of record at four companies. Works in TypeScript, Go, and Python on PostgreSQL, BigQuery, and Snowflake, with LLM-powered features and workflow orchestration in production. On-call at every role.

EXPERIENCE

Full-Stack ML / AI Engineer, Live-Commerce Startup

2025 - Present

- Engineered multi-tenant backend services and REST/gRPC APIs in TypeScript, Go, and Python on AlloyDB (PostgreSQL) and ClickHouse, with rate limiting, caching, and load balancing, handling **100K requests/sec** for analytics and real-time personalization
- Built customer data integrations syncing product data from external commerce platforms, with warehouse transformations managed in Dataform
- Designed event-driven inference infrastructure on GCP serving foundation models (vLLM, RunPod serverless GPUs) for LLM-powered features, using Pub/Sub queues, workflow orchestration, automatic retries, and KEDA autoscaling for traffic spikes
- Instrumented services with OpenTelemetry, Cloud Monitoring, and Prometheus alerting, sustaining **99.9%+ availability** on real-time APIs since launch

MLOps / Cloud Engineer, Jabil (Badger Technologies)

2023 - 2025

- Designed and operated Apache Spark pipelines ingesting telemetry from a fleet of **35,000 robots**, each producing ~25 GB/day of 8K camera imagery and diagnostic data, with orchestrated workflows and automatic recovery for failed jobs
- Built lakehouse storage and BigQuery and Snowflake warehouse layers with versioned datasets, fed by batch and streaming pipelines on Kafka and Pub/Sub and by integrations with customers' external systems
- Owned a multi-tenant data and inference platform across AWS, GCP, and Azure (Kubernetes, Terraform, Helm), maintaining **99.99% availability** as part of the on-call rotation
- Shipped LLM-powered features and production AI models with evaluation and tracing to measure output quality and catch regressions
- Built monitoring services in Go and TypeScript (Prometheus, Grafana, New Relic) with latency and throughput SLO alerting, improving system performance **15%**

DevOps / Cloud Engineer, Amazon Web Services (AWS)

2022 - 2023

- Operated AWS data services for customer workloads (Redshift, Athena, EMR/Spark, Kinesis, Glue, S3), building workflow orchestration for batch and streaming pipelines and ingestion from customers' external systems

- Automated infrastructure with Terraform, CDKTF (TypeScript), and Kubernetes, cutting deployment times **20%** through GitHub Actions CI/CD
- Designed tiered monitoring with Prometheus, Grafana, and OpenTelemetry, reducing production downtime **15%**; served on the on-call rotation and standardized post-mortems to reduce MTTR

Software Engineer, Air Liquide / Airgas

2021 - 2022

- Engineered Apache Spark indexing and ETL pipelines for product catalog and search data spanning **millions of SKUs**, powering downstream search and recommendation services
 - Integrated product data from external systems into the catalog pipeline
 - Built microservices and low-latency APIs in TypeScript, Go, Java, and Python, automated with Terraform, Docker, Ansible, and GitLab CI/CD
 - Improved search query response time **15%** through bottleneck analysis, caching, and index optimization
-

SKILLS

Languages: TypeScript, Go, Python, Java, SQL

Databases and Warehouses: PostgreSQL, AlloyDB, BigQuery, Snowflake, Redshift, Athena, ClickHouse, MongoDB

Data Engineering: Apache Spark, ETL/ELT, batch and streaming pipelines, Kafka, Pub/Sub, Dagster, Dataform, Iceberg, Delta Lake

Backend and Distributed Systems: REST APIs, gRPC, microservices, multi-tenant SaaS, event-driven architecture, queues and workers, retries and failure recovery

AI and ML: LLM-powered features, model serving (vLLM), evaluation and tracing, AWS Bedrock, MLOps

Cloud and Infrastructure: GCP, AWS, Azure, Kubernetes, Terraform, CDKTF, Helm, ArgoCD, Docker, GitHub Actions

Observability: OpenTelemetry, Prometheus, Grafana, New Relic, on-call, post-mortems

EDUCATION

B.S. Computer Science, University of Phoenix